Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/12/2012
PI Name:	Bhattacharjee, Subrata Ph.D.		
Project Title:	Residence Time Driven Flame Spread Over Solid Fuels		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	prof.bhattacharjee@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	619-594-6080
Organization Name:	San Diego State University		
PI Address 1:	5500 Campanile Drive, Mechanical Engineering Department		
PI Address 2:			
PI Web Page:			
City:	San Diego	**State:**	CA
Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	6
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	16
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2014, per NSSC information (Ed., 9/17/2013) NOTE: Extended to 12/31/2013, per NSSC information (Ed., 12/31/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Miller, Fletcher (San Diego State University) Paolini, Christopher (San Diego State University) Takahashi, Shuhei (Gifu University) Wakai, Kazunori (Gifu University)		
Grant/Contract No.:	NNX10AE03G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Flame spread over solid fuels in an opposed-flow environment has been investigated for over four decades for understanding the fundamental nature of hazardous fire spread. The appeal for this configuration stems from the fact that flame spread rate remains steady, even if the flame itself may grow in size. For practical fire safety issues, however, wind-assisted flame spread is more relevant. However, these two regimes have always been studied in isolation without much effort to establish a connection, even though the underlying mechanism of flame spread is the same in all regimes. Sitting between the two regimes are high-residence time flames, as found in a low-velocity or quiescent microgravity environment. Residence time is the time spent by an oxidizer in the combustion zone. Such flames, which are of interest on their own merit due to fire safety issues in spacecrafts, offer some unique characteristics because of the high residence time. Radiation becomes dominant and, based on previous space experiments and analysis, we contend that a vigorously spreading flame on Earth becomes self-extinguishing in a microgravity environment under certain conditions such as the fuel thickness being greater than a critical value. The proposed research uses a comprehensive approach-- a novel experimental set up and a theoretical framework based on scaling and numerical modeling-- to investigate flame spread driven by varying residence time, from blow-off extinction in an opposed-flow configuration through high residence time flame to blow-off extinction in a concurrent-flow configuration. At the heart of this proposal is a novel but simple experiment where the residence time of the oxidizer can be controlled and high residence time flames can be established for a long duration (compared to drop towers). As a proof of concept, we have constructed a flame tower at SDSU in which, after a sample is ignited, the sample holder, placed in an open moveable cart, can be traversed at any desired speed upward or downward, creating an external flow that can augment or mitigate the buoyancy-induced flow. Preliminary results show that we can control the residence time and create flames in different regimes, including a transition between a wind-aided and wind-opposed configuration. At Gifu University in Japan, we have been developing an interferometry based imaging system which we intend to enhance to capture the thermal footprint of a flame's leading edge. The leading edge is central to our understanding of mechanism of flame extinction. Further development of this technology will enable us to integrate diagnostics in future space based experiments and provide validation data to a comprehensive numerical model. The comprehensive model, to be built upon our existing two-dimensional model, will solve an unsteady, three-dimensional, Navier stokes equation with finite rate kinetics in the gas and solid phases and radiation in the gas phase. The software implementation will be object-oriented and utilize a new technology called Web Services that will decouple various sub-models and enhance parallel execution. The radiation model will also be refined by including the equilibrium composition of species for finding radiative properties in high residence-time flames. The new data generated by our proposed ground-based experiment will be the first of its kind to encompass flame spread across different regimes and will stimulate fundamental theoretical research to unify different regimes of flame spread. The comprehensive model, tested against available theory, data in literature, and data generated at SDSU and Gifu, will be applied to test the three hypotheses presented in this proposal regarding flame extinguishment in a microgravity environment. A successful outcome of this project will lead to a well thought out space-based experiment on the mechanism of flame extinction in a gravity free environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research has three components: We have built a novel experimental facility called Flame Tower at SDSU, where ground based experiments on flame spread are being conducted; Theoretical and computational work that explore the similarity and differences between the mechanisms flame spread in a zero gravity space environment and on Earth; Design and propose a space based experiment to establish extinction mechanism of flames. The data that we are acquiring in the Flame Tower will provide the research community with a comprehensive set of results for testing different theories of flame spread in a normal gravity environment. Moreover, by controlling the residence time, various regimes of flame spread, including the microgravity regime, can be explored in the Flame Tower. Our theoretical work predicts a fuel thickness beyond which steady flame spread is unsustainable in a gravity free environment. If we are successful in establishing a critical thickness, this will have a powerful impact on making fire resistant environment for humans in space.
	1. Abstract This has been the most productive period for this project. We have made significant progress on several fronts: i) establishing the flame stabilizer as a novel apparatus for flame spread study by presenting it to the international community of combustion researchers at the 34th International Symposium on Combustion, ii) Acquiring data on flame spread rate and flame shape in the SDSU flame tower, constructed and calibrated over past two years, iii) numerical modeling of downward spreading flame and comparing the computed results with experiments for both cellulose and PMMA for flame spread and for flame shape, iv) development of an expression for effective opposed-flow velocity to correlate blow-of extinction with the developing boundary layer upstream, and v) refining the objectives and requirements for the space based experiments. 2. Executive Summary of original proposed research Flame spread over solid fuels in an opposed-flow environment has been investigated for over four decades for understanding the fundamental nature of hazardous fire spread. The appeal for this configuration stems from the fact that the opposed-flow flame spread rate remains steady, even if the flame itself may grow in size. For practical fire safety issues, however, wind-assisted flame spread is more relevant. However, these two regimes have always been studied in isolation without much effort to establish a connection, even though the underlying mechanism of flame spread is the same in all regimes. Sitting between the two regimes are high-residence time flames, as found in a low-velocity or quiescent microgravity environment. Residence time is the time spent by an oxidizer in the combustion zone. Such flames, which are of interest in their own merit due to fire safety issues in spacecrafts, offer some unique characteristics because of the high residence time. Radiation becomes dominant and, based on previous space experiments and analysis, we contend that a vigorously spreading flame on earth becomes self-extinguishing in a microgravity environment under certain conditions such as the fuel thickness being greater than a critical value. The proposed research uses a comprehensive approach – a novel experimental set up and a theoretical framework based on scaling and numerical modeling – to investigate flame spread driven by varying residence time, from blow-off extinction in an opposed-flow configuration through high residence time to blow-off extinction in a concurrent-flow

Task Progress:

configuration.

At the heart of this research is a novel but simple experiment where the residence time of the oxidizer is controlled and high residence time flames are established for a long duration (compared to drop towers). As a proof of concept, we have constructed a flame tower at SDSU in which, after a sample is ignited, the sample holder, placed in an open moveable cart, can be traversed at any desired speed upward or downward, creating an external flow that can augment or mitigate the buoyancy-induced flow. Preliminary results show that we can control the residence time and create flames in different regimes, including a transition between a wind-aided and wind-opposed configuration. At Gifu University in Japan, we have been developing an interferometry based imaging system that we intend to enhance to capture the thermal footprint of a flame's leading edge. The leading edge is central to our understanding of the mechanism of flame extinction. Further development of this technology will enable us to integrate diagnostics in future space based experiments and provide validation data to a comprehensive numerical model. We will build the comprehensive model upon our existing two-dimensional model that will solve an unsteady, three-dimensional, Navier-Stokes equation with finite rate kinetics in the gas and solid phases and radiation in the gas phase. The software implementation will be object-oriented and will utilize a new technology called Web Services that will decouple various sub-models and enhance parallel execution. We will refine the radiation model by including the equilibrium composition of species for finding radiative properties in high residence-time flames.

The new data generated by our proposed ground-based experiment will be the first of its kind to encompass flame spread across different regimes and will stimulate fundamental theoretical research to unify different regimes of flame spread. We will apply the comprehensive model, tested against available theory, data in literature, and data generated at SDSU and Gifu, to test the three hypotheses presented in this proposal regarding flame extinguishment in a microgravity environment. A successful outcome of this project will lead to a well thought space-based experiment on the mechanism of flame extinction in a gravity free environment.

3. Ground based Experimental Approach

3.1: SDSU Flame Tower: The flame tower is the centerpiece of our ground-based activities. We have finished the construction of a 10m tall steel chamber (details on year-1 report) inside which a fuel sample mounted on a cart can be traversed up or down a vertical rail with a prescribed velocity. We have been successful in developing a completely remote controlled system to move a cart at any desired speed (from -2 m/s to + 2 m/s: details in year-2 report). We have conducted detailed velocity measurement to establish that the flow seen by the flame is uniform upstream over a 40 mm by 40 mm area upstream of the fuel sample (which is 20 mm wide).

The design and operation of the micro flame tracker, which is housed inside the moving cart, has been described in the year-2 report. Once the flame is ignited, a gas phase thermocouple, attached to a linear motion system on the cart, tracks the motion of the leading edge of the flame, providing the instantaneous flame spread rate. The flame spread is also obtained by analyzing the side-view digital video of the flame, allowing us to verify the data from the automated tracking system.

Data on flame spread rate and flame shape were obtained for flame spread over ash less filter paper with the relative flow velocity varying from positive (opposed flow configuration) to negative (concurrent flow configuration). The spread rate behavior was consistent with theoretical prediction for the opposed flow configuration. When the cart was moved upward (in the same direction of the buoyancy driven flow), the flame spread rate remained fairly constant (or slightly increasing) until about a flow speed of -40 cm/s, when the flame converted itself into its concurrent-flow configuration (wind assisted flame spread). We are still analyzing the wealth of data produced by these experiments. The flow velocity at which blow-off extinction occurs was found to be sensitive to the development length of the boundary layer. In order to minimize the effect of the sample holder, we are redesigning the micro flame tracker so that the sample sees nothing but the flow of oxidizer. We also are in the process of acquiring data with thin films of PMMA.

3.2: Flame Stabilizer: One of the challenges in the experimental study of flame spread is that, even if the flame spreads at a steady rate, the propagating flame creates an unsteady phenomenon with respect to the laboratory frame of reference. As a result, it is difficult to obtain detailed data, necessary for validating models, in a spreading flame. To remedy this situation, we have built a novel flame spread apparatus that moves the fuel in the opposite direction of the flame spread to keep the leading edge of the flame stationary with respect to the laboratory. A thermocouple, fixed to the laboratory frame of reference, in front of the leading edge of the flame senses the presence of the flame and a PID controller keeps its temperature constant by moving the sample holder, driven by a stepper motor, in the opposite direction at the velocity of the spread. Instantaneous flame spread rate and the visible flame structure are compared for a downward spreading flame over ashless filter paper with the corresponding stationary flame. The results indicate that the difference between the two configurations are within experimental uncertainties and the stabilized flame can represent a spreading flame adequately, including variability of flame spread rate and the flame geometry, for further observation. We have presented this work in the 34th International Symposium on Combustion. With a spreading flame stabilized by this apparatus, we are in a position to measure gas phase temperature, including in the plume region, where fluctuations due to turbulence makes it very difficult to map out the thermal field of a spreading flame. We are also making CO2 field measurements and comparing the gas phase fields with our computational results.

4. Space based Experimental Approach

We have refined the objectives of our space-based experiments and produced a concept design for the proposed experiment. The objectives for the space-based experiments are the following: a. Establish that steady flame spread over solid fuel in a quiescent microgravity environment is possible only if the fuel thickness is less than a critical value. b. Investigate the propagation of the products and temperature fields to uncover the mechanism of flame extinguishment. c. Determine how the ambient pressure and presence of a mild opposing flow affect the flame spread rate and extinction behavior.

The basic concept of the proposed experiment is as follows: A carousal driven by a stepper motor holds several sample holders, each of which are placed in the test zone by rotating the carousal. Samples of different thicknesses are held in each sample holder, which carries its own igniter. When a sample is rotated into the test zone, the igniter snaps into the ignition circuit. As the flame spreads over the sample, gas phase thermocouples and CO2 sensors monitor the propagation of the thermal diffusion fronts. Once the propagating flame reaches the end of the sample or extinguishes, the recording of the temperature and species field will continue to provide valuable data on transient cooling of the gas phase and the diffusion of species. After a preset interval, the next sample is rotated into position and the experiment is repeated for a new fuel thickness. The carousel will then be replaced with a second carousel and the ambient pressure

will be reduced, which is predicted to be a significant parameter just like the fuel thickness in determining the critical fuel thickness.

We are using the flame stabilizer to probe a standing flame with CO2 probe and thermocouples to see their impact on the flame. The purpose is to ascertain that the diagnostics of the proposed experiment do not distort the flame being observed.

5. Theoretical/Modeling Work

We have made considerable progress in modeling flame spread over solid fuels under different conditions. Our modeling/theoretical effort can be summarized as follows:

5.1. We have been developing Web based tools for calculating equilibrium temperature of PMMA and Cellulose combustion. This calculation tools, which can be used by the community, helps us determine exactly how much sample burn is possible (under different conditions) in a closed chamber without significantly altering the oxygen level. It also predicts the equilibrium composition providing us with the thermodynamic limits of CO2 level and temperature in the gas phase to be expected.

5.2. We are using a two dimensional model with finite-rate one-step kinetics, and radiation to simulate opposed flow flame spread. The model has been used to compare downward flame spread results with experiments conducted in the lab. The spread rate from the model for three different fuel thicknesses agreed quite well with the experimental results for downward spread over PMMA sheets under the ambient conditions.

5.3. The model was used to compare pure downward flame spread with the stabilized flame produced by our stabilizer device. The comparison of the numerical results as well as experimental data established the flame stabilizer does reproduce all the characteristics of a downward spreading flame, only the flame is now frozen in the laboratory coordinate ready for prolonged examination. This study established the flame stabilizer as a new viable platform for experimental studies of flame spread.

5.4. The data from the flame tower showed that the blow off velocity (of the opposing flow) was related to the boundary layer development length. The computational model, along with scale analysis, was used to quantify the effect of the development length in terms of an effective velocity. An effective velocity for a flame, embedded in a boundary layer, is defined as an equivalent velocity seen by the flame. The effective velocity is then correlated with free stream velocity, development length of the boundary layer, and fluid and fuel properties. The resulting correlations were remarkably accurate in explaining the blow off extinction velocity over a wide range of parametric conditions.

5.5 Work is underway to obtain the detailed radiation field inside the flame so that thermocouple measurements can be corrected for radiation loss or gain for a spreading flame.

Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Bhattacharjee S, Paolini C, Patterson M. "A web service infrastructure and its application for distributed chemical equilibrium computation." Journal of Computational Science Education. 2012 Jun;3(1):19-27. http://www.shodor.org/media/content//jocse/volume3/issue1/Bhattacharjee , Jun-2012
Articles in Peer-reviewed Journals	Bhattacharjee S, Bundy M, Paolini C, Patel G, Tran W. "A novel apparatus for flame spread study." Proceedings of the Combustion Institute. 2013;34(2):2513-21. http://dx.doi.org/10.1016/j.proci.2012.05.076 , Jan-2013
Articles in Peer-reviewed Journals	Bhattacharjee S, Paolini C, Miller F, Nagarkar R. "Radiation signature in opposed-flow flame spread." Progress in Computational Fluid Dynamics. 2012;12(4):293-301. http://dx.doi.org/10.1504/PCFD.2012.048249 , May-2012
Articles in Peer-reviewed Journals	Paolini C, Bhattacharjee S. "IGE Model: An extension of the ideal gas model to include chemical composition as part of the equilibrium state." Journal of Thermodynamics. 2012:Article ID 870631, 18 p. http://dx.doi.org/10.1155/2012/870631 , Mar-2012
Papers from Meeting Proceedings	Bhattacharjee S, Paolini C, Kumar M. "A Learning Outcome Driven Cyber Infrastructure for Thermodynamics Education." XSEDE12 First Conference of the Extreme Science and Engineering Discovery Environment, Chicago, IL, July 16-19, 2012. XSEDE12 First Conference of the Extreme Science and Engineering Discovery Environment, Chicago, IL, July 16-19, 2012. Paper #127 , Jul-2012
Papers from Meeting Proceedings	Tran W, Paolini C, Bhattacharjee S. "Measuring Flame Temperature." Third Access Outreach, San Diego State University, March 2012. Third Access Outreach, San Diego State University, March 2012. , Mar-2012
Papers from Meeting Proceedings	Bhattacharjee S, Al Ghamdi A, Tran W, Miller F, Paolini C. "Measurement of Temperature Field in a Stabilized Downward Spreading Flame." Western States Section Meeting of the Combustion Institute, Arizona State University, Marh 19-20, 2012. Western States Section Meeting of the Combustion Institute, Arizona State University, March 19-20, 2012. Paper No. 12S-07. , Mar-2012
Papers from Meeting Proceedings	Paolini C, Bhattacharjee S. "An Equilibrium Approach to Modeling O2-H2 Combustion Kinetics." Western States Section Meeting of the Combustion Institute, Arizona State University, March 19-20, 2012. Western States Section Meeting of the Combustion Institute, Arizona State University, March 19-20, 2012. Paper No. 12S-36. , Mar-2012